Assignment 1

Team Members

1. Mariam Ashraf Hassan Mohamed
2. Hager Essam Gaber Mahmoud
3. Rama Ehab Ahmed Said
4. Seif Ashraf Fawzy Roushdy

1. Introduction & Project Motivation

- **Project Name:** AiSentinel Dashboard.
- **Concept:** An AI-powered cybersecurity platform designed to detect, analyze, and respond to threats in real time. The system combines rule-based detection with machine learning models to identify known attacks, detect anomalies, and automate incident response through a unified SOC dashboard.
- **Problem Statement:** Traditional dashboards are overwhelmed with data, leading to "Alert Fatigue" for security analysts.
- **Goal:** To simplify security monitoring and reduce manual intervention through intelligent automation.

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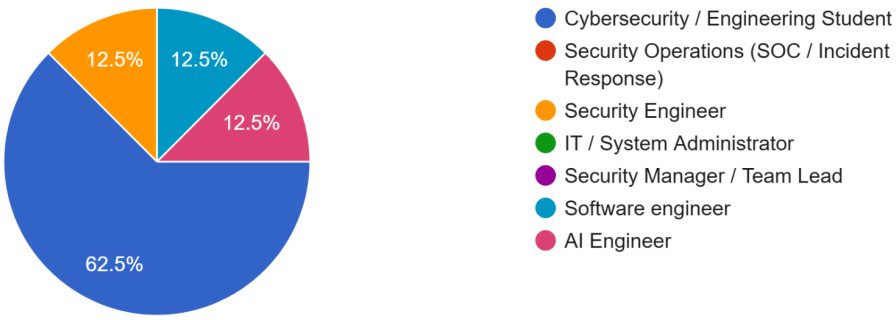
2. User Research Methodology

- **Research Method:** Via an Online Survey (Google Forms).
- **Target Audience:** Cybersecurity Professionals (SOC Analysts, Security Engineers) and IT/Software Engineering Students.
- **Sample Size:** (8 Participants).

What is your current role?

8 responses



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3. Empathize Phase:

Based on the survey responses, we have gathered critical data to understand the challenges and expectations of our users.

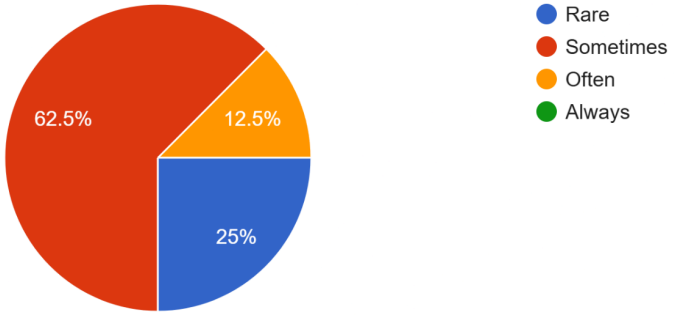
A. User Experience & Challenges (Pains)

- **Alert Fatigue:** The majority of users face alert fatigue "Sometimes" to "Often," highlighting a clear need for noise reduction.

- **Key Frustrations:** The most significant pain point is the **"Difficulty in distinguishing real threats from false positives,"** followed closely by a **"Lack of automation in response"**.
- **Current Tools:** Users currently rely on a mix of **Custom/In-house Dashboards**, **Splunk**, and basic tools like **Windows Defender**, suggesting a gap for a more unified, intelligent solution.

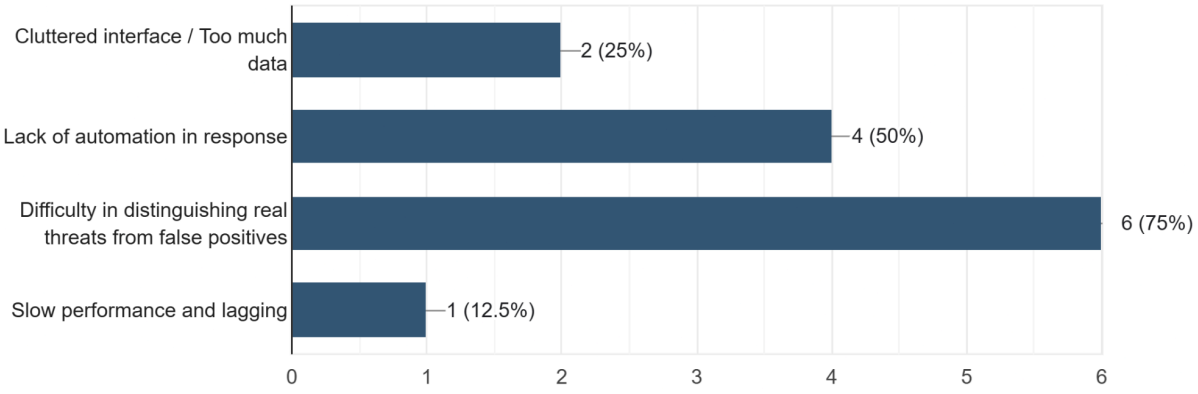
How often do you face alert fatigue?

8 responses



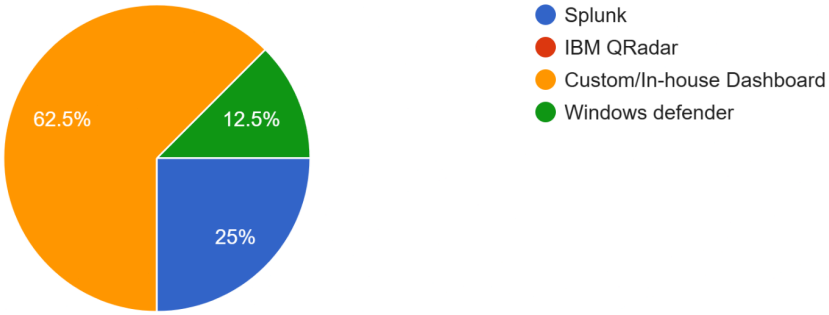
What is the most frustrating part of current security dashboards?

8 responses



What tools do you currently use for monitoring security?

8 responses

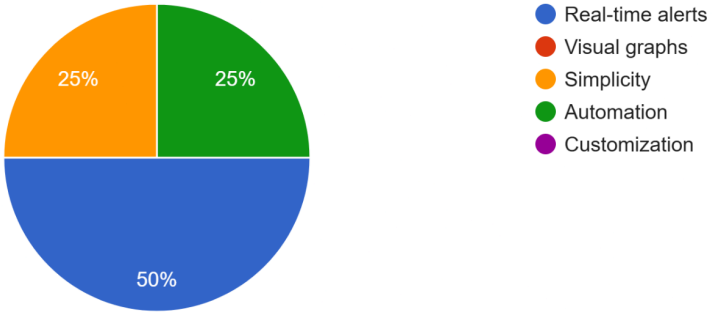


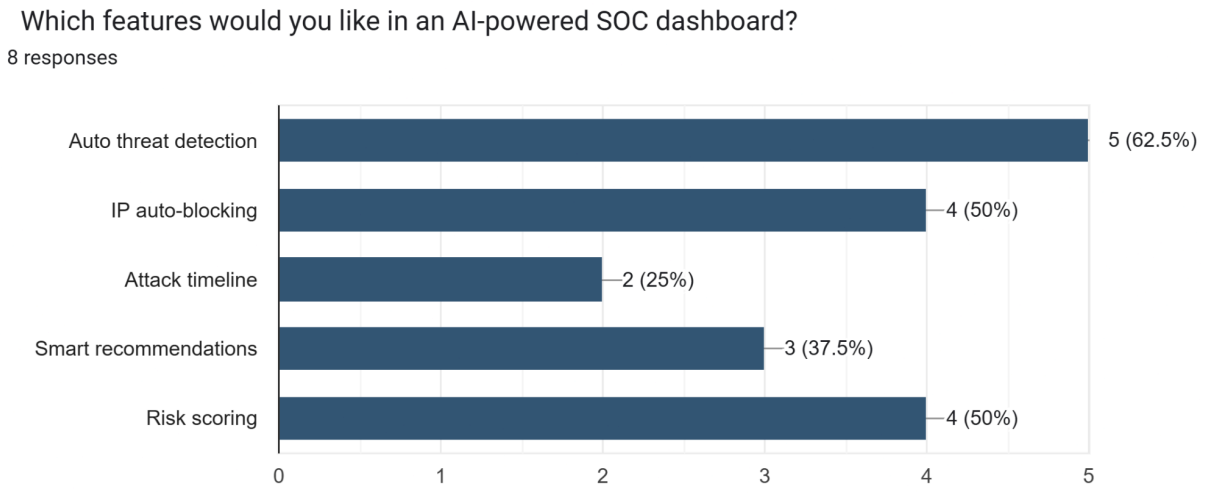
B. User Needs & Gains

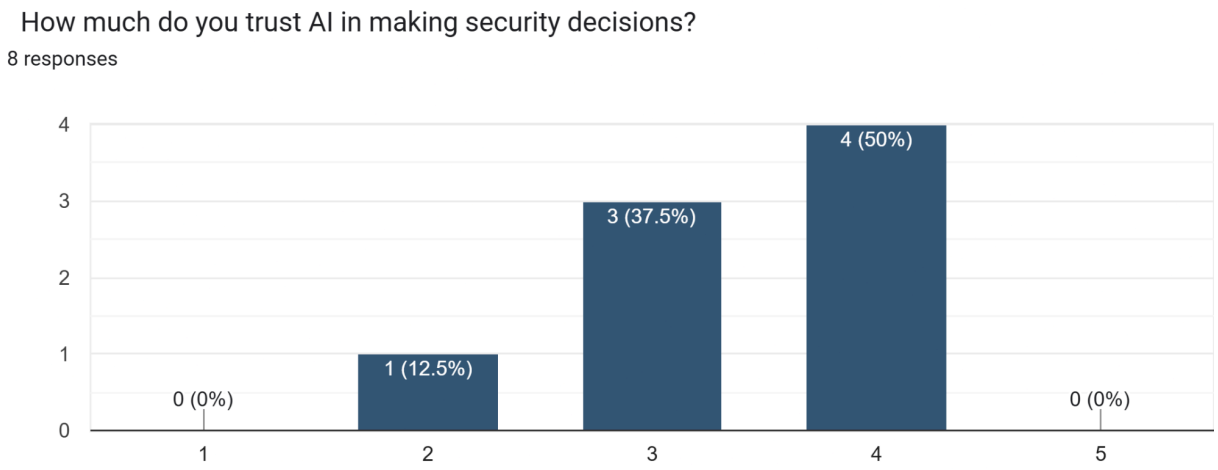
- **Critical Features: "Real-time alerts"** and **"Automation"** were ranked as the most important aspects of a security dashboard.
- **AI Integration:** Users are highly interested in **"Auto threat detection," "IP auto-blocking,"** and **"Risk scoring"**.
- **Trust in AI:** On a scale of 1 to 5, the average trust in AI for security decisions is approximately **3.5**, indicating that users are open to AI but prefer it for high-risk automation rather than full control.

What matters most in a security dashboard?

8 responses







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4. User Personas

Based on the data, we identified two primary user segments for **AiSentinel**:

Persona 1: The Expert Defender

- **Name:** Shahd (Security Engineer)
- **Background:** Has extensive experience with security monitoring systems like Splunk.
- **Goals:** Wants a **Customizable** layout with **Real-time alerts** and **IP auto-blocking** to speed up incident response.
- **Frustration:** The lack of automation in current response workflows.

Persona 2: The Aspiring Specialist

- **Name:** Hagar (Cybersecurity Student)
- **Background:** Occasionally uses custom dashboards and is learning threat patterns.
- **Goals:** Needs a **Minimal** or **Customizable** interface with **Smart recommendations** and **Attack timelines** to understand threats better.
- **Frustration:** Cluttered interfaces and difficulty distinguishing real threats from noise.

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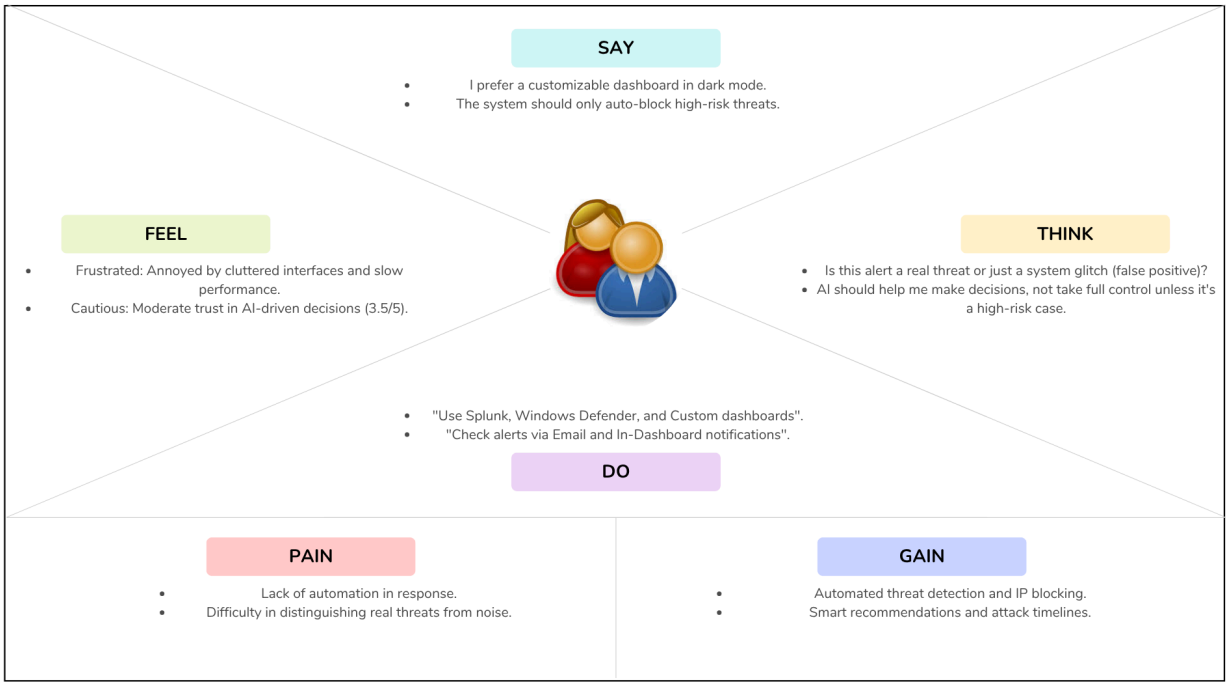
5. Empathy Map

This map summarizes the collective mindset of our participants:

- **Says**: "I prefer a customizable dashboard in dark mode" and "The system should only auto-block high-risk threats".
- **Thinks**: "I need a system that helps me prioritize alerts so I don't feel overwhelmed".
- **Does**: Switches between different monitoring tools and checks email or in-dashboard notifications for alerts.
- **Feels**: Frustrated by slow performance and the high volume of false positives in current systems.



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6. Conclusion & Selection Reason

We selected the **AiSentinel Dashboard** project because our research confirms that:

1. **Market Need:** Even experienced engineers and students struggle with "Alert Fatigue" and "False Positives".
2. **Solution Validation:** There is a high demand for AI-driven features like **Auto Threat Detection** and **Risk Scoring**.
3. **UI/UX Opportunity:** Most users prefer **Dark Mode** and **Customizable layouts**, which our project will prioritize to enhance the overall user experience.

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7. Links

- AiSentinel Dashboard - User Survey (Responses)
  AiSentinel Dashboard - User Survey (Responses)
- AiSentinel Dashboard - User Survey
  https://forms.gle/HUnFugJq9QGZVw1c6

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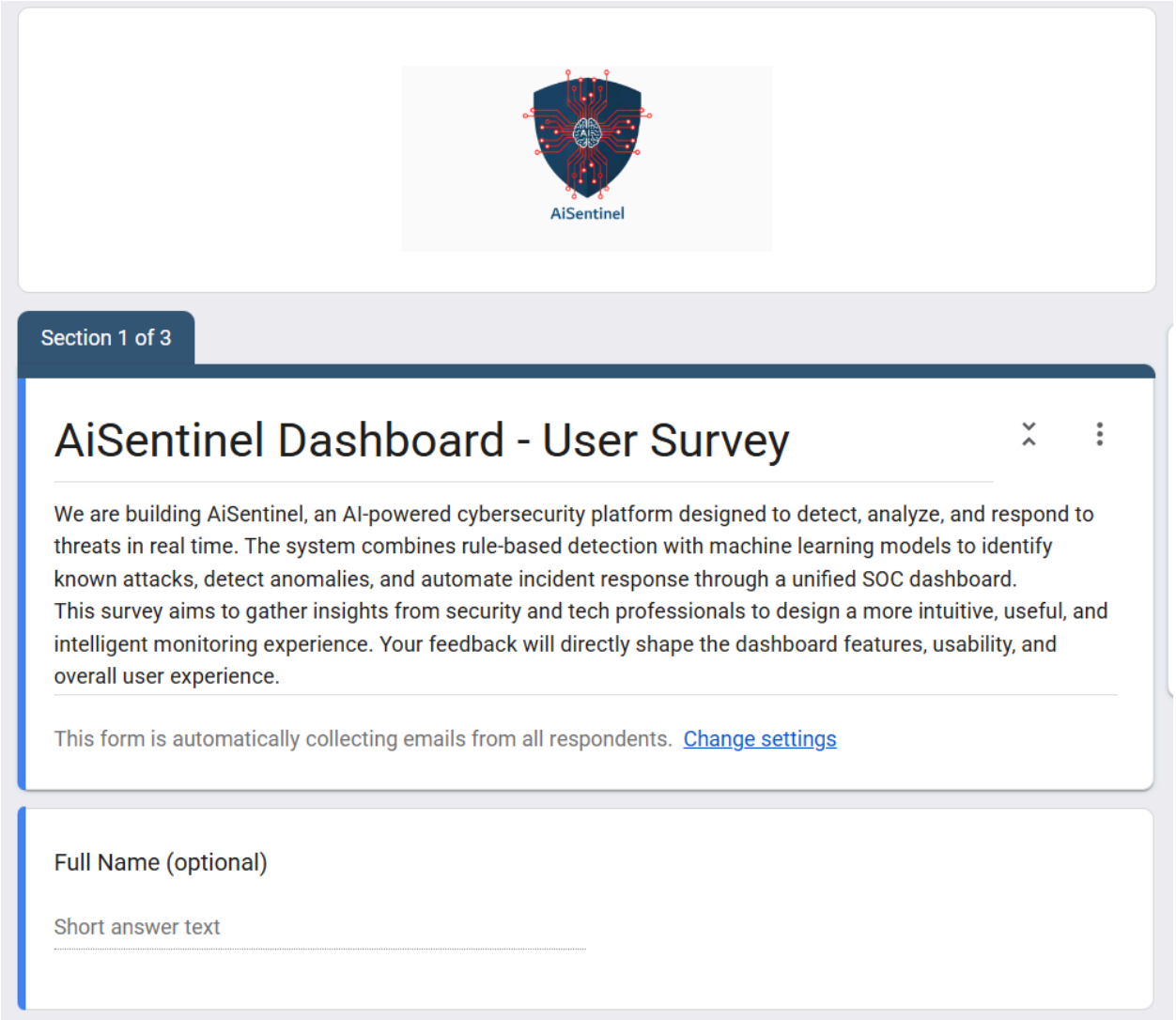
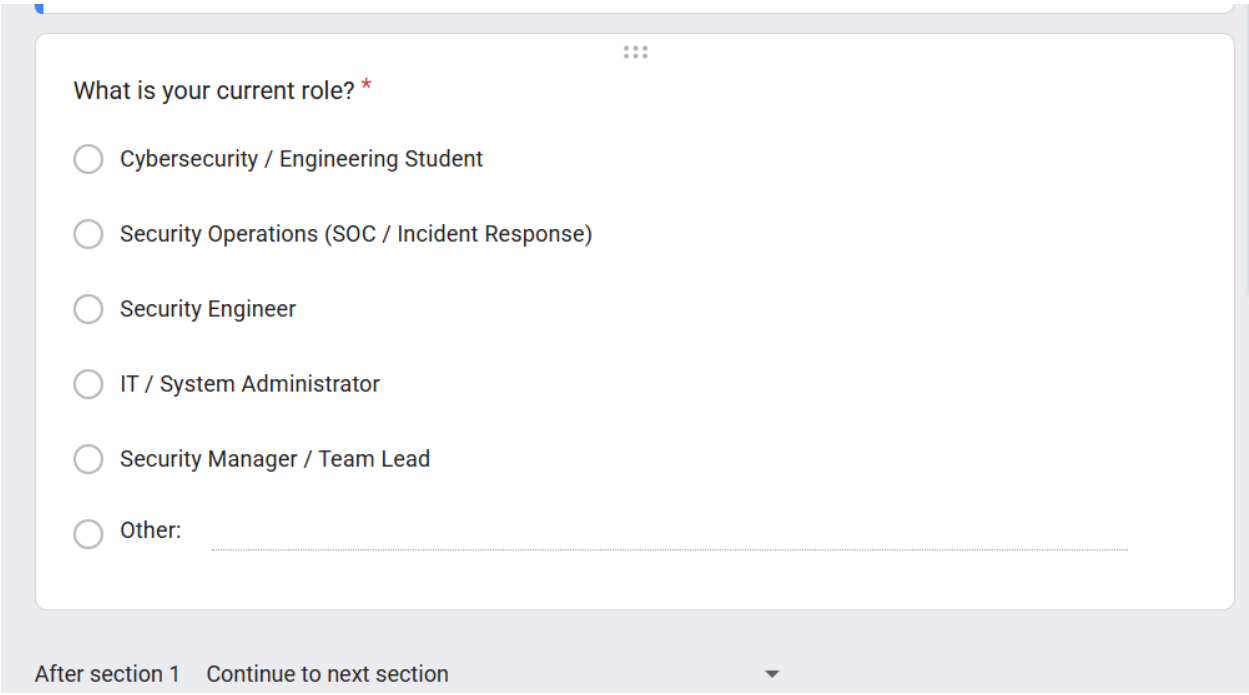
8. Screenshots





Section 2 of 3

Section title (optional)

Description (optional)

Have you ever used security monitoring systems before? *

- Yes, extensively
- Yes, but only occasionally or in a limited way
- No, I haven't

What tools do you currently use for monitoring security? *

- Splunk
- IBM QRadar
- Custom/In-house Dashboard
- Other: _____

How often do you face alert fatigue? *

- Rare
- Sometimes
- Often
- Always

What is the most frustrating part of current security dashboards? *

- [ ] Cluttered interface / Too much data
- [ ] Lack of automation in response
- [ ] Difficulty in distinguishing real threats from false positives
- [ ] Slow performance and lagging
- [ ] Other: _____

After section 2 Continue to next section

Section 3 of 3

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Description (optional)

What matters most in a security dashboard? *

- Real-time alerts
- Visual graphs
- Simplicity
- Automation
- Customization

Which features would you like in an AI-powered SOC dashboard? *

- [ ] Auto threat detection
- [ ] IP auto-blocking
- [ ] Attack timeline
- [ ] Smart recommendations
- [ ] Risk scoring

How do you prefer receiving alerts? *

- ( ) In-Dashboard notifications only
- ( ) Email notifications
- ( ) Instant messaging (Slack / Microsoft Teams)
- ( ) Telegram Bot alerts

Should the system auto-block attackers? *

- ( ) Yes always
- ( ) Only high-risk
- ( ) No (manual only)

How much do you trust AI in making security decisions? *

	1	2	3	4	5	
Not at all	( )	( )	( )	( )	( )	Fully trust

What type of dashboard layout do you prefer? *

- Minimal
- Detailed
- Customizable

Do you prefer dark mode? *

- Yes
- No
- Optional

Any other thoughts or feedback to help us improve the AiSentinel Dashboard ?

Long answer text

Who has responded?

Email
hager20121999@gmail.com
hagar.khaled485@gmail.com
roaa71829@gmail.com
noor.alhoda.eisa@gmail.com
hassanadel0115@gmail.com
mariam.ashraf.eng@gmail.com
hageresam513@gmail.com
d.shahdahmed020@gmail.com

Full Name (optional)

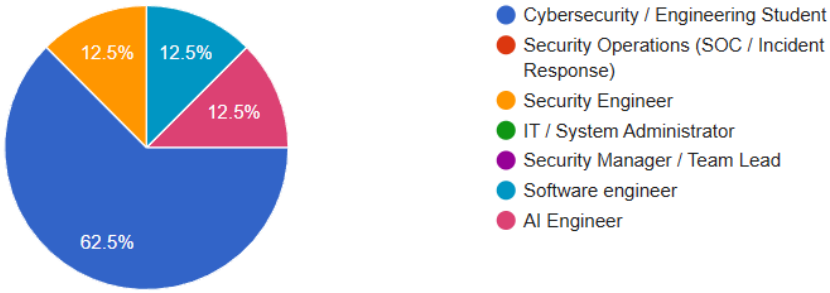
4 responses

- shahd
- رؤى محمود عبد الوهاب
- Hagar
- HAGAR

What is your current role?

Copy chart

8 responses



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What tools do you currently use for monitoring security?

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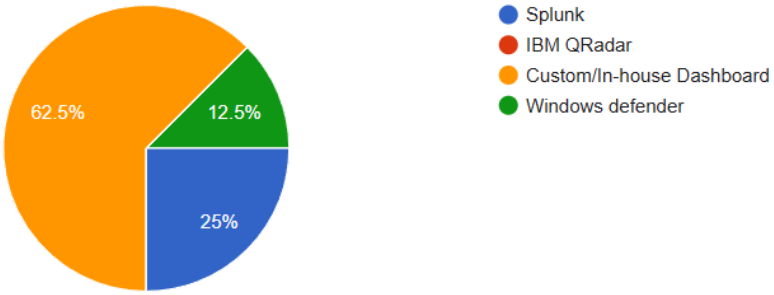

How often do you face alert fatigue?

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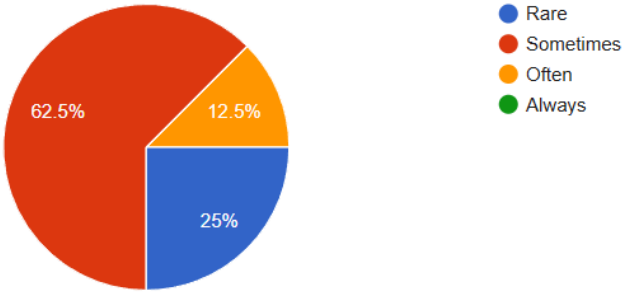

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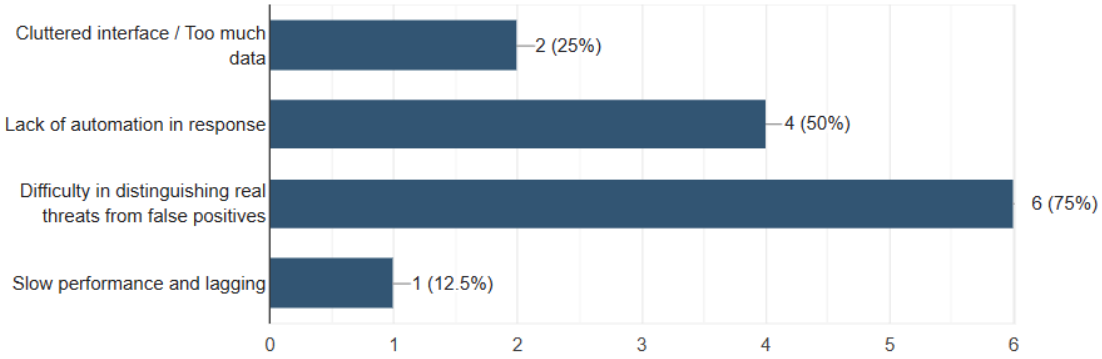

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What matters most in a security dashboard?

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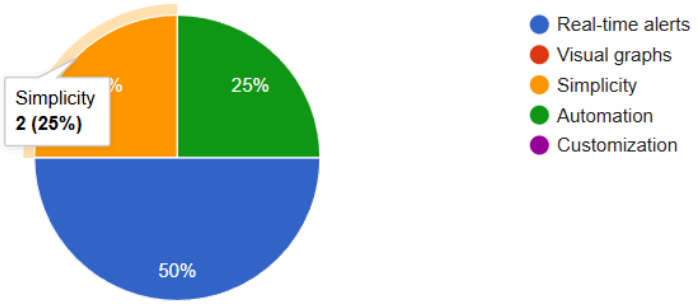
8 responses



Which features would you like in an AI-powered SOC dashboard?

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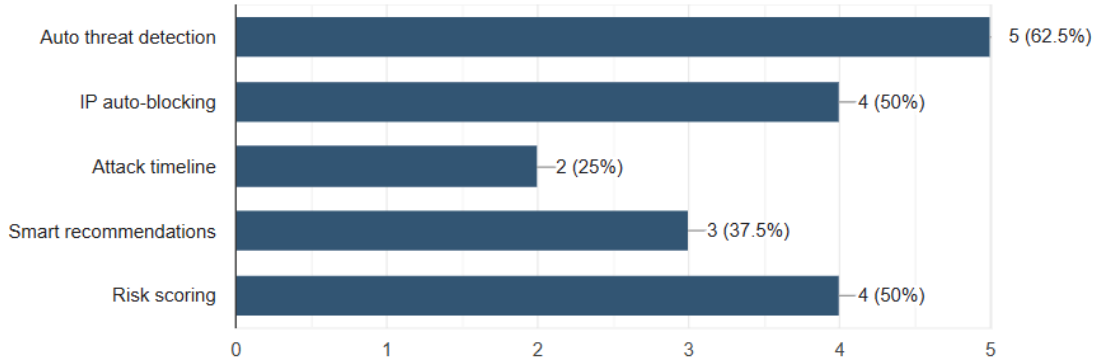
8 responses



How do you prefer receiving alerts?

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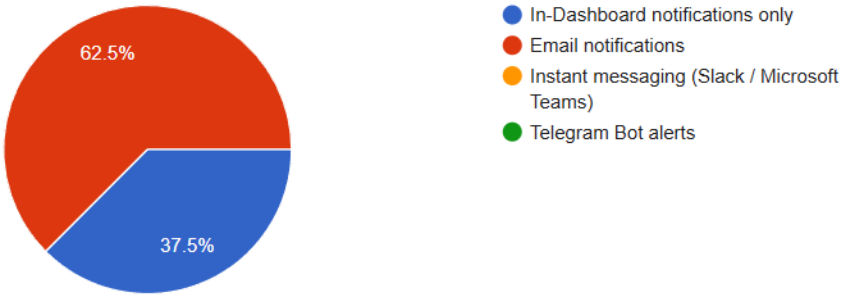
8 responses



Should the system auto-block attackers?

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8 responses

How much do you trust AI in making security decisions?

Copy chart

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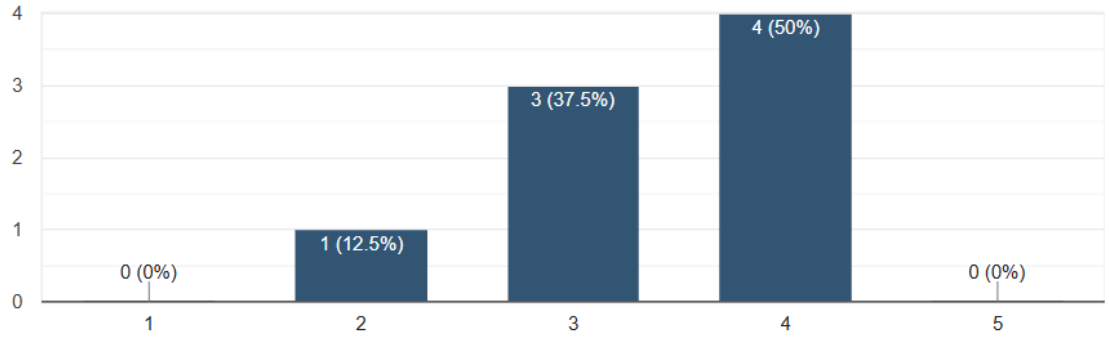
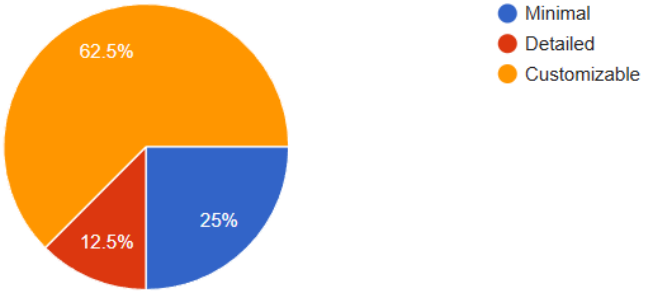


What type of dashboard layout do you prefer?

Copy chart

8 responses



Do you prefer dark mode?

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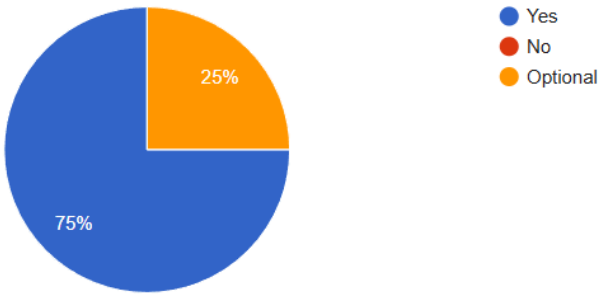
8 responses



Any other thoughts or feedback to help us improve the AiSentinel Dashboard ?

0 responses

No responses yet for this question.